

A Cross-OSDR Network-Contrast Framework for Kidney Spaceflight Aging

Working methodology memo

RRRM-2 Kidney Transcriptome Project

May 2026

Abstract

This memo translates the current project debate into a concrete analysis plan. The goal is not to replace the original network-rewiring question with a generic WGCNA story. The goal is to keep the original estimand—whether spaceflight amplifies, dampens, reverses, or redirects age-associated kidney network change—while moving from a single underpowered dataset to a cross-OSDR network-contrast framework. RRRM-2 remains the only exact Age $\times$ Arm $\times$ Flight kidney RNA-seq design. Other OSDR kidney datasets cannot replicate the full factorial question, but they can test whether RRRM-2 flight network directions, node-level scores, module-level programs, and silent-shifter candidates recur across independent missions, strains, sexes, durations, and platforms.

Contents

1	Plain-English Summary	3
2	What This Is Not	3
3	How We Got Here	4
3.1	Starting point: age-dependent network rewiring	4
3.2	First constraint: separate condition networks are not credible	5
3.3	Second move: WGCNA as a stability and annotation layer	5
3.4	Third move: LIONESS and sample-specific networks	5
3.5	Fourth move: silent shifters	6
3.6	Fifth move: node2vec and latent topology	6
3.7	Sixth move: the grey60/WGCNA convergence attempt	6
3.8	Seventh move: OSD-253 as a proposed TLR4 discriminator	7
3.9	Current synthesis: contrast-vector decomposition	7
4	Scientific Question	7
4.1	Original question	7
4.2	What WGCNA can and cannot answer	8
5	Dataset Roles	8
6	Core Notation	9
6.1	Residualized expression	9
6.2	Network vector	10

7	Backbone Strategy	10
7.1	Why a shared backbone is needed	10
7.2	Frozen candidate edge universe	10
8	RRRM-2: Exact Primary Estimands	10
8.1	Control aging network vector	11
8.2	Flight aging network vector	11
8.3	Spaceflight modification of age rewiring	11
9	Amplify, Dampen, Reverse, Redirect	11
9.1	Projection coefficient	11
9.2	Cosine similarity	11
9.3	Orthogonal / redirected component	12
10	Inference	12
10.1	Permutation tests	12
10.2	Avoiding overclaiming	12
11	Cross-OSDR Extension	13
11.1	Within-study flight network vectors	13
11.2	Cross-study projection tests	13
11.3	Meta-analysis	13
12	Silent Shifters	14
12.1	Within-study score	14
12.2	Cross-study criterion	14
13	Role of WGCNA and Modules	14
14	Tabula Muris Senis and Aging References	14
15	Metadata-Specific Rules	15
15.1	OSD-253	15
15.2	OSD-771	15
16	Proposed Analysis Phases	15
17	Minimum Viable Version	17
18	Bottom Line	17

1 Plain-English Summary

The original project should not be thrown away. The original idea is still the right scientific question:

Does spaceflight alter age-related kidney transcriptional network organization, and does it do so differently in ISS-T and LAR?

The problem is that RRRM-2 alone has $n = 5$ per Age $\times$ Arm $\times$ Flight cell. That is enough to estimate some module- or pathway-scale effects, but it is not enough to make individual edge discovery carry the paper. Therefore the framework should shift from:

single-dataset edge discovery

to:

multi-cohort network contrast estimation, projection, and replication at edge, gene, module, and pathway resolution.

The key idea is simple: a contrast can be represented as a vector. The vector can be an expression vector, but for this project it should primarily be a *network* vector:

- an edge-effect vector;
- a gene differential-connectivity vector;
- a centrality-change vector;
- a delta-kME vector;
- a LIONESS-derived node-score vector;
- or a module/pathway activity vector.

Then the question “amplifies, dampens, or redirects” becomes a geometry question. If the flight-aging network vector points in the same direction as the control-aging network vector, flight is aging-like or amplifying. If it points in the opposite direction, flight is reversing. If most of it is orthogonal, flight is redirecting the kidney into a different network state.

2 What This Is Not

This plan is not:

- a generic WGCNA paper;
- a DE-only or pathway-only paper;
- a claim that Tabula Muris Senis droplet data should be used to build raw co-expression networks;
- a raw pooled mega-analysis that ignores mission, strain, sex, platform, duration, or dissection protocol;
- another attempt to force $n = 5$ separate condition-level correlation networks to produce stable individual edge discoveries.

This plan is:

- a way to preserve the original network-rewiring question;
- a way to use RRRM-2 as the only exact factorial bridge;
- a way to use other OSDR kidney datasets as independent flight-network replication or context cohorts;
- a way to decide which claims survive beyond RRRM-2.

3 How We Got Here

This section is included because the current proposal can sound, at first, like a sudden pivot into dot products or embeddings. It is not. It is the result of walking the original question through several methods and keeping only the parts that still answer the biology honestly.

3.1 Starting point: age-dependent network rewiring

The original project was not “do differential expression and then WGCNA.” The original project was:

Use the RRRM-2 factorial design to ask whether spaceflight changes the way mouse kidney transcriptional networks age, separately for ISS-T and LAR, after accounting for kidney cell composition.

That implies several desired outputs:

- estimate age-associated network change in controls;
- estimate age-associated network change under flight;
- test whether flight amplifies, dampens, reverses, or redirects the aging network direction;
- do this separately for ISS-T and LAR;
- find genes with high network change but modest differential expression;
- interpret those genes with targeted kidney biology, not generic pathway lists;
- validate signals with sample-specific networks, external cohorts, or metadata-aware projections.

In statistical terms, the original estimand was always a contrast of contrasts:

$$(\text{Old}_{FLT} - \text{Young}_{FLT}) - (\text{Old}_{GC} - \text{Young}_{GC}).$$

For this project, the object inside each parenthesis is not only expression. It can be a network edge, node score, module score, pathway score, or sample-specific network summary.

3.2 First constraint: separate condition networks are not credible

The most direct version of the idea would build one network for each condition: Young GC, Old GC, Young FLT, Old FLT, separately inside ISS-T and LAR. That would match the biology neatly, but RRRM-2 has only about $n = 5$ per cell. At that size, condition-specific correlations are too unstable to be the main inferential object. This forced the first methodological move:

Do not estimate completely separate networks per condition. Estimate effects on a shared or constrained network coordinate system.

That is why shared-topology networks, fixed edge universes, shrinkage correlations, differential connectivity summaries, and LIONESS-style sample-specific networks entered the project.

3.3 Second move: WGCNA as a stability and annotation layer

WGCNA entered for a sensible reason. If individual edges are unstable, modules can provide a more stable biological resolution. WGCNA can define co-expression modules, eigengenes, hub genes, and module-trait associations. That is useful.

The problem is that WGCNA does not naturally answer the original factorial network question. Its module eigengene is the first principal component of a module’s expression matrix. That eigengene can detect a Flight $\times$ Age or Flight $\times$ Arm pattern if the relevant signal lies on the dominant module axis. But if the interaction lives in a smaller-variance within-module direction, the eigengene can blur it away. Also, WGCNA module-trait association is not the same as edge-level or node-level rewiring.

Therefore WGCNA was demoted from:

primary proof of age-dependent rewiring

to:

secondary structure, compression, and biological annotation.

That demotion is not anti-WGCNA. It is matching the method to the question.

3.4 Third move: LIONESS and sample-specific networks

LIONESS entered because the project needed sample-level network summaries. A sample-specific network gives each mouse a network feature vector, which can then be modeled with ordinary factorial machinery:

$$\text{network feature} \sim \text{Age} \times \text{Flight} \times \text{Arm} + \text{covariates}.$$

This is closer to the original design than a single pooled WGCNA eigengene test. It lets the model ask whether a network feature changes with age differently under flight and whether that differs between ISS-T and LAR.

The cost is multiple testing. Edge-wise LIONESS models across many edges are underpowered at RRRM-2 sample size, so strict FDR can be sparse or null. That does not make the method wrong. It means individual edge discovery cannot carry the manuscript alone.

3.5 Fourth move: silent shifters

The silent-shifter idea was created to capture a biologically plausible pattern:

Some genes may not change much in mean expression, but their network neighborhood, connectivity, or regulatory context may change substantially.

That is still a good idea. The issue is that a single-study silent-shifter list can be fragile. A stronger version treats silent shifting as a ranking or recurrence phenomenon:

genes repeatedly high in network-change rank and low in differential-expression rank across strata or studies.

So silent shifters remain useful, but as reproducible node-level candidates and pathway/module annotations rather than as a hard single-study gene list.

3.6 Fifth move: node2vec and latent topology

Node2vec entered for the same reason silent shifters entered: it seemed like a way to measure whether a gene’s network neighborhood moved even when expression did not. In principle, if a gene shifts from one network neighborhood to another, its learned graph embedding can move.

The problem is interpretability. A node2vec coordinate is a latent coordinate. Its axes are not fixed biological quantities, embeddings from different graphs must be aligned, and results can depend on graph construction, random-walk parameters, embedding dimension, and random seed. That makes node2vec reasonable as an exploratory description of topology movement, but weak as the primary factorial estimand.

This is why the current plan avoids saying “make embeddings and compare them.” The preferred language is:

estimate interpretable network contrast vectors, then decompose their direction, magnitude, and orthogonal redirected component.

Node2vec can still be used as a sensitivity or exploratory layer, but not as the main claim.

3.7 Sixth move: the grey60/WGCNA convergence attempt

One attempted rescue was to make WGCNA grey60 the biological center of the paper and ask whether LIONESS-factorial genes converged into that module. The repository results did not support that claim. The 11 ISS-T-Young LIONESS candidate genes did not concentrate in grey60. The best overlap was marginal in purple, and grey60 had zero overlap with those candidates.

At the eigengene level, grey60 did show an ISS-T-specific pattern, but so did royalblue, salmon, and pink. Therefore the correct inference is not:

one unique grey60 ECM module drives the entire response.

The more honest inference is:

RRRM-2 contains an ISS-T-biased module-level signal across several biological programs, while gene-level network rankings are method-dependent.

That result was important because it prevented the manuscript from becoming a WGCNA-first story that no longer matched the original question.

3.8 Seventh move: OSD-253 as a proposed TLR4 discriminator

OSD-253 was considered as a possible validation or mechanism test because it contains C57BL/6J and C3H/HeJ mice, and C3H/HeJ carries impaired TLR4 signaling. If grey60 were uniquely TLR4-dependent, OSD-253 might have shown selective attenuation of grey60-like module projection in C3H/HeJ.

The metadata made this more complicated. OSD-253 has original controls and rerun controls with light wavelength, read length, and run-context differences. Therefore it cannot be used casually as a clean TLR4 validation. A metadata-aware projection is useful, but only as scenario-based context. The current repository analysis did not provide a clean grey60-specific C3H attenuation result.

This means OSD-253 should not decide the whole paper. It should constrain and contextualize mechanistic claims.

3.9 Current synthesis: contrast-vector decomposition

The current proposal is the cleaned-up version of the original methodology. It keeps the original biological question but changes the level at which the statistics are expected to be reliable.

Earlier version	Current version
Find individual rewired edges in RRRM-2.	Estimate global, module, pathway, and node-level network contrast vectors first; treat individual edges as exploratory.
Use WGCNA as primary evidence.	Use WGCNA as module definition, compression, and annotation.
Use node2vec as central topology movement metric.	Use node2vec only as exploratory; prefer interpretable contrast vectors.
Validate one grey60 mechanism in OSD-253.	Use OSD-253 as metadata-aware strain/duration context, not a clean decisive validation.
RRRM-2 must succeed alone.	RRRM-2 defines the exact factorial contrast; external OSDR cohorts test whether flight-network directions recur.

So the current framework is not a retreat from the original project. It is the same project expressed at a statistically defensible resolution.

4 Scientific Question

4.1 Original question

The original question can be written as:

After accounting for kidney cell composition, does spaceflight amplify, dampen, reverse, or redirect age-associated kidney transcriptional network change, separately in ISS-T and LAR?

The key feature is that this is a network contrast question, not simply a module eigengene association question.

4.2 What WGCNA can and cannot answer

WGCNA is useful for defining stable co-expression modules, module eigengenes, hub genes, and module-trait associations. It can support statements such as:

This module is flight-associated, age-associated, or enriched for DCT, ECM, ER stress, transport, immune, or mitochondrial biology.

But WGCNA is not the native method for asking whether the age-related *network contrast* changes under flight. Standard module eigengene models can miss genes whose average expression is stable but whose network neighborhood changes. Therefore WGCNA should be a secondary annotation and compression layer, not the primary estimand.

5 Dataset Roles

Dataset	Assay / tissue	Design role	Use in this framework
OSD-771 GLDS-674	/ Kidney bulk RNA-seq	RRRM-2. Female C57BL/6NTac. Spaceflight, age, euthanasia location / arm. Young and old; ISS-T and LAR; FLT, GC, VIV, BSL.	Primary exact Age $\times$ Arm $\times$ Flight network-contrast dataset.
OSD-513 GLDS-513	/ Kidney bulk RNA-seq	RR-23. Male C57BL/6J. 20 FLT, 20 HGC, 20 VGC; no age factor.	Best powered flight network-response anchor. Validate whether RRRM-2 flight network directions recur in a larger cohort.
OSD-253 GLDS-253	/ Kidney bulk RNA-seq	RR-7. Female C57BL/6J and C3H/HeJ. 25 and 75 day durations. Original GC and GC rerun controls have important light/read-length complications.	Strain and duration context. Do not treat as a clean TLR4 validation without metadata-aware sensitivity analyses.
OSD-102 GLDS-102	/ Kidney bulk RNA-seq plus multi-omics	RR-1. Female C57BL/6J, 16 weeks, FLT versus GC. No age factor.	Flight network context and replication of module/pathway/node directions.

Dataset	Assay / tissue	Design role	Use in this framework
OSD-163 GLDS-163	/ Kidney bulk RNA-seq plus multi-omics	RR-3. Female BALB/c, 12 weeks, FLT versus GC. No age factor.	Strain/context replication. Useful for whether flight network programs are C57-specific or broader.
OSD-462 GLDS-462	/ Kidney transcriptomics, proteomics, phosphoproteomics	RR-10. Female WT / p21-null design; WT subset can be used carefully. No age factor.	Mechanistic context for stress, cell-cycle, p21, and proteomic support.
OSD-457 GLDS-457	/ Multi-tissue RNA-seq including kidney	JAXA MHU-3. WT and Nrf2-KO, spaceflight and genotype. No age factor.	Stress-response and genotype context, especially Nrf2/oxidative stress.
OSD-913 GLDS-735	/ Kidney miRNA-seq	RRRM-1 / RR-8. Age, flight, euthanasia location, dissection condition. Not bulk mRNA.	Regulatory-layer age $\times$ flight context. Do not use as mRNA co-expression network evidence.
Tabula Muris Senis / aging references	Single-cell or bulk aging atlases	Independent aging references.	Use for deconvolution, cell-type aging signatures, and annotation. Do not build raw droplet co-expression networks as the primary network reference.

6 Core Notation

Let X_s be the residualized expression matrix for study s , with genes $g \in G$ and samples $i \in I_s$.

6.1 Residualized expression

Within each study, residualize expression using metadata-appropriate covariates:

$$X_{gi} = \mu_g + C_i \alpha_g + T_i \gamma_g + \varepsilon_{gi}$$

where C_i includes estimated kidney cell-type composition and T_i includes technical covariates such as batch, library batch, RIN/RINe, read length, sequencing run, and study-specific confounds.

The residuals $\hat{\varepsilon}_{gi}$ are the input for network estimation.

6.2 Network vector

Define a frozen edge universe $E = \{(g, h)\}$ on a common gene panel. For a contrast c , define an edge-effect vector:

$$\mathbf{e}_{s,c} = (\Delta w_{s,c,1}, \Delta w_{s,c,2}, \dots, \Delta w_{s,c,|E|})$$

where $\Delta w_{s,c,k}$ is the contrast effect for edge k in study s .

Also define gene-level node vectors:

$$\mathbf{n}_{s,c} = (\Delta N_{s,c,1}, \Delta N_{s,c,2}, \dots, \Delta N_{s,c,|G|})$$

where ΔN can be differential connectivity, centrality change, delta-kME, or an aggregated LIONESS node score.

7 Backbone Strategy

7.1 Why a shared backbone is needed

Separate $n = 5$ correlation networks are unstable. A raw Pearson correlation has Fisher- z standard error approximately

$$\text{SE}(z) = \frac{1}{\sqrt{n-3}}.$$

At $n = 5$, this is about 0.71, before even taking differences between conditions. Therefore condition-specific networks cannot be the primary object.

7.2 Frozen candidate edge universe

Use a frozen candidate edge universe rather than testing all gene pairs. Options:

1. **Control-only shrinkage backbone:** build a Ledoit-Wolf / shrinkage partial-correlation backbone from pooled residualized controls, excluding FLT labels from edge selection.
2. **Multi-study control backbone:** build study-specific control backbones, then take the union or recurring top- k edges.
3. **Biology-constrained backbone:** restrict to edges within or between kidney-relevant pathways/modules, such as DCT/transport, ECM, ER stress, mitochondria, immune, lipid metabolism, and tubular injury.

The backbone is not itself the inferential result. It is the coordinate system for contrast estimation.

8 RRRM-2: Exact Primary Estimands

RRRM-2 is the only exact dataset for the original question. Define network vectors separately by arm.

8.1 Control aging network vector

For arm $a \in \{\text{ISS-T}, \text{LAR}\}$:

$$\mathbf{A}_{GC}^a = \mathbf{N}(\text{Old}, a, GC) - \mathbf{N}(\text{Young}, a, GC)$$

This is the age-associated network change in controls.

8.2 Flight aging network vector

$$\mathbf{A}_{FLT}^a = \mathbf{N}(\text{Old}, a, FLT) - \mathbf{N}(\text{Young}, a, FLT)$$

This is the age-associated network difference under flight.

8.3 Spaceflight modification of age rewiring

$$\Delta_{AgeRewire}^a = \mathbf{A}_{FLT}^a - \mathbf{A}_{GC}^a$$

This is the within-arm Flight $\times$ Age network interaction. The arm difference is:

$$\Delta_{Arm} = \Delta_{AgeRewire}^{ISS-T} - \Delta_{AgeRewire}^{LAR}.$$

9 Amplify, Dampen, Reverse, Redirect

This is where the dot product enters. The dot product is not replacing network analysis; it is a way to classify the direction of one network contrast relative to another.

9.1 Projection coefficient

For a given arm:

$$\beta_a = \frac{\mathbf{A}_{FLT}^a \cdot \mathbf{A}_{GC}^a}{\mathbf{A}_{GC}^a \cdot \mathbf{A}_{GC}^a}.$$

Interpretation:

- $\beta_a > 1$: flight amplifies the control aging network direction.
- $0 < \beta_a < 1$: flight dampens the control aging network direction.
- $\beta_a < 0$: flight reverses the control aging network direction.
- $\beta_a \approx 0$: flight is mostly not aligned with the control aging network direction.

9.2 Cosine similarity

$$\cos(\theta_a) = \frac{\mathbf{A}_{FLT}^a \cdot \mathbf{A}_{GC}^a}{\|\mathbf{A}_{FLT}^a\| \|\mathbf{A}_{GC}^a\|}.$$

This measures directional similarity independent of magnitude.

9.3 Orthogonal / redirected component

$$\mathbf{R}_a = \mathbf{A}_{FLT}^a - \beta_a \mathbf{A}_{GC}^a.$$

$$\rho_a = \frac{\|\mathbf{R}_a\|}{\|\mathbf{A}_{FLT}^a\|}.$$

Interpretation:

- small ρ_a : flight aging pattern is mostly along the control aging axis;
- large ρ_a : flight aging pattern is redirected into a different network direction.

10 Inference

10.1 Permutation tests

For RRRM-2, use label permutations that respect the design:

- permute FLT/GC labels within Age $\times$ Arm when testing flight effects;
- permute Young/Old labels within Arm $\times$ Environment when testing age effects;
- preserve cage/batch constraints where metadata requires it;
- recompute contrast vectors, projections, and orthogonal fractions under each permutation.

Primary permutation statistics:

- β_a for amplification/dampening;
- $\cos(\theta_a)$ for directional alignment;
- ρ_a for redirectedness;
- $\|\Delta_{AgeRewire}^a\|$ for interaction magnitude;
- module/pathway concentration of $\mathbf{R}_a$.

10.2 Avoiding overclaiming

Individual edge FDR should not be the sole success criterion. At RRRM-2 cell sizes, edge-wise FDR is expected to be sparse or null. The hierarchy should be:

1. global network-vector statistic;
2. module/pathway concentration;
3. gene/node ranking;
4. individual edges only as exploratory annotations.

11 Cross-OSDR Extension

Other OSDR studies do not generally have age $\times$ arm structure. Therefore they cannot replicate the full RRRM-2 estimand. They can replicate pieces.

11.1 Within-study flight network vectors

For each external study s , define:

$$\mathbf{F}_s = \mathbf{N}_s(FLT) - \mathbf{N}_s(GC/HGC).$$

Where possible, define study-specific variants:

- OSD-253: $\mathbf{F}_{strain,duration,control}$ for strain, duration, and original versus rerun controls;
- OSD-513: powered male C57 flight vector;
- OSD-102: female C57 flight vector;
- OSD-163: BALB/c flight vector;
- OSD-462: WT subset flight vector;
- OSD-457: WT and Nrf2-KO flight vectors if sample sizes permit.

11.2 Cross-study projection tests

Test whether external flight network vectors align with RRRM-2-derived vectors:

$$\cos(\mathbf{F}_s, \mathbf{F}_{RRRM2,ISS})$$

$$\cos(\mathbf{F}_s, \Delta_{AgeRewire}^{ISS-T})$$

$$\cos(\mathbf{F}_s, \mathbf{R}_{ISS})$$

This asks whether the RRRM-2 network direction is recurring across missions, not whether every external dataset has the same factorial design.

11.3 Meta-analysis

For gene/node-level scores, combine study-level signed statistics using:

- signed Stouffer Z with weights proportional to effective sample size;
- random-effects meta-analysis for module/pathway scores;
- rank aggregation for silent-shifter candidates;
- leave-one-study-out robustness checks.

Do not pool raw expression across missions as the primary analysis. Pool contrast-level statistics.

12 Silent Shifters

The single-study silent-shifter definition is fragile. In the cross-OSDR framework, silent shifters should become a reproducibility concept.

12.1 Within-study score

For each gene g :

$$S_{g,s} = \text{network_change}_{g,s}$$

$$D_{g,s} = |\text{DE}_{g,s}|$$

A gene is a candidate silent shifter in study s if it has high $S_{g,s}$ and low $D_{g,s}$.

12.2 Cross-study criterion

A stronger definition:

A reproducible silent shifter is a gene with consistently high network-change rank across studies or strata, while remaining modest in differential-expression rank.

This should be tested with enrichment and rank aggregation, not treated as a single-study gene list with strong individual-gene claims.

13 Role of WGCNA and Modules

WGCNA should remain in the project, but its role should be explicit:

- define or annotate stable residual-expression modules;
- compress network and expression signals into module-level summaries;
- ask whether high network-vector loadings concentrate in modules;
- interpret redirected components biologically;
- test whether RRRM-2 module signals project into external cohorts.

WGCNA should not be asked to prove age-dependent topology rewiring.

14 Tabula Muris Senis and Aging References

Tabula Muris Senis should not be used as a raw droplet co-expression network source for this paper. It can be used in three safer ways:

1. **Cell-type deconvolution reference:** the current use case.
2. **Cell-type-specific aging annotations:** pseudobulk within cell type and age where metadata supports animal-level aggregation.

3. **Biological labels:** identify whether network-loaded genes are PT, DCT, LOH, collecting duct, endothelial, immune, fibroblast, or other cell-type-biased aging genes.

If a good independent bulk kidney aging reference is available, use it to provide an external aging axis. But that should be secondary to the network aging axis estimated inside RRRM-2.

15 Metadata-Specific Rules

15.1 OSD-253

OSD-253 requires separate scenarios:

- original GC: matched read length, but FLT is white-light and original GC is blue-light;
- white-light rerun GC: fixes light wavelength, but changes read length and run context;
- blue-light rerun GC: sensitivity only.

Any strain/TLR4 statement must survive metadata-aware scenario analysis.

15.2 OSD-771

OSD-771 is the primary factorial bridge. Euthanasia location / arm must be treated as part of the biological and procedural design:

- ISS-T FLT: terminal on-orbit collection;
- LAR FLT: return to Earth and recovery;
- controls are Earth-collected;
- therefore ISS-T-specific signals cannot be casually called purely “acute flight” without discussing collection protocol.

16 Proposed Analysis Phases

Phase 0: Dataset inventory

- Download or verify VST/count matrices and ISA metadata for OSD-771, OSD-513, OSD-253, OSD-102, OSD-163, OSD-462, OSD-457, and OSD-913.
- Create a harmonized metadata table with study, strain, sex, age, duration, FLT/GC/VIV/BSL, arm/euthanasia, platform, read length, library batch, RIN, and control scenario.

Phase 1: Harmonization and residualization

- Map genes to common Ensembl IDs.
- Compute cell-type deconvolution using the existing kidney reference.
- Fit study-specific residualization models.
- Save residual matrices with provenance.

Phase 2: Frozen backbone

- Define one or more frozen edge universes.
- Record exactly how edges were selected.
- Lock the backbone before contrast interpretation.

Phase 3: Within-study network contrasts

- RRRM-2: estimate full Age $\times$ Arm $\times$ Flight network contrasts.
- OSD-513 and other cohorts: estimate FLT versus GC/HGC network vectors.
- OSD-253: estimate strain-, duration-, and control-scenario-specific network vectors.

Phase 4: Projection and redirection

- Compute β , cosine similarity, and orthogonal fraction for RRRM-2.
- Compute cross-study alignment of external flight vectors with RRRM-2 network vectors.
- Use permutation or bootstrap for uncertainty.

Phase 5: Gene, module, and pathway interpretation

- Aggregate edges to genes.
- Identify reproducible node-level contributors.
- Test module/pathway enrichment of aligned and redirected components.
- Re-evaluate silent shifters as recurring rank patterns.

Phase 6: Decision

Observed result	Manuscript implication
RRRM-2 age-rewiring vector is significant and recurs across external flight vectors.	Strong network-aging paper. Spaceflight modifies kidney aging network architecture.
RRRM-2 is weak alone, but external flight vectors align with RRRM-2 flight or redirected network programs.	Cross-OSDR flight-network biology paper. RRRM-2 provides the factorial bridge; external cohorts provide recurrence.
Only WGCNA module activity is positive; network vectors do not recur.	Do not force a network-aging paper. Consider a modest module-activity paper or stop.

Observed result	Manuscript implication
No coherent module, pathway, node, or projection signal survives.	Do not write RRRM-2 as a standalone biological network paper. Use as preliminary or negative constraint evidence.

17 Minimum Viable Version

If time is limited, the minimum viable version is:

1. RRRM-2 exact network-vector redirection test using the existing residualized/LIONESS/direct-correlation machinery.
2. OSD-513 powered FLT versus HGC network vector.
3. OSD-253 metadata-aware strain/duration sensitivity.
4. Compare vector alignment and module/pathway concentration across these three.

This is enough to determine whether the cross-OSDR network paper is alive.

18 Bottom Line

The original methodology is not being discarded. It is being reframed from:

“RRRM-2 alone discovers age-dependent network rewiring genes.”

to:

“RRRM-2 defines the only exact age-by-arm flight network contrast, and a cross-OSDR framework tests whether that network direction recurs across independent kidney space-flight cohorts.”

That is closer to the original ambition than a WGCNA-only paper, and it avoids pretending that $n = 5$ per factorial cell can support strong individual edge discovery by itself.

Useful Source Links

- OSD-771 / RRRM-2 kidney RNA-seq: <https://osdr.nasa.gov/bio/repo/data/studies/OSD-771>
- OSD-513 / RR-23 kidney RNA-seq: <https://osdr.nasa.gov/bio/repo/data/studies/OSD-513>
- OSD-253 / RR-7 kidney RNA-seq: <https://osdr.nasa.gov/bio/repo/data/studies/OSD-253>
- OSD-102 / RR-1 kidney: <https://osdr.nasa.gov/bio/repo/data/studies/OSD-102>
- OSD-163 / RR-3 kidney: <https://osdr.nasa.gov/bio/repo/data/studies/OSD-163>
- OSD-462 / RR-10 kidney: <https://osdr.nasa.gov/bio/repo/data/studies/OSD-462>

- OSD-457 / MHU-3 multi-tissue: <https://osdr.nasa.gov/bio/repo/data/studies/OSD-457>
- OSD-913 / RRRM-1 kidney miRNA: <https://osdr.nasa.gov/bio/repo/data/studies/OSD-913>